🏆 Infosys SE Elite Crack Vault — 01 Number Systems LCM HCF and Divisibility Rules Elite

🎯 Infosys Systems Engineer (SE) | 📊 9 Questions | 📅 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

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Problem 01 • Number Systems & Simplification

🔥 MUST-SOLVE

`ID: SEC02_BOOK_P05_Q29` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> $$1 \div \frac{3}{7} \text{ of } (6 + 8 \times 3 - 2) + \left[\frac{1}{5} \div \frac{7}{25} - \left{ \frac{3}{7} + \frac{8}{14} \right} \right]$$

- a. $-\frac{2}{7}$
- b. $-\frac{3}{9}$
- c. $-\frac{7}{3}$
- d. $-\frac{5}{42}$

✓ **Verified Answer:** `D`

Problem 02 • Number Systems & Simplification

🔥 MUST-SOLVE

`ID: SEC02_BOOK_P07_Q46` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> How many numbers between 400 and 1000 can be formed with the digits 0, 2, 3, 4, 5, 6, if no digit is repeated in the same number?

- a. 40
- b. 50
- c. 60
- d. 70

✓ **Verified Answer:** `C`

Problem 03 • Number Systems & Simplification

MUST-SOLVE

`ID: SEC02_BOOK_P07_Q47` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> How many numbers between 300 and 3,000 can be formed with the digits 0, 1, 2, 3, 4, 5 (without repetition)?

- a. 60
- b. 180
- c. 300
- d. 420

✓ **Verified Answer:** `B`

Problem 04 • Number Systems & Simplification

MUST-SOLVE

`ID: SEC02_BOOK_P07_Q48` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> How many 3-digit numbers can be formed from the digits 2, 3, 5, 6, 7 and 9, which are divisible by 5 and none of the digits is repeated?

- a. 5
- b. 10
- c. 15
- d. 20

✓ **Verified Answer:** `D`

Problem 05 • Number Systems & Simplification

CORE SE

`ID: SEC02_BOOK_P07_Q49` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> How many even four-digit numbers can be formed with the digits 0, 1, 2, 3, 4, 5, 6; no digit being used more than once?

- a. 100
- b. 269
- c. 342
- d. 420

✓ **Verified Answer:** `D`

Problem 06 • Number Systems & Simplification

CORE SE

`ID: SEC02_BOOK_P08_Q50` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> MR. X has forgotten his friend's seven-digit telephone number. He remembers the following: the first three digits are either 635 or 674, the number is odd, and the number nine appears once. If X were to use a trial and error process to reach his friend, what is the number of trials he has to make before he can be certain to succeed?

- a. 1000
- b. 3402
- c. 2430
- d. 3006

✓ **Verified Answer:** `B`

Problem 07 • Number Systems & Simplification

CORE SE

`ID: SEC02_BOOK_P08_Q51` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> A Four-digit number is formed using digits 1, 2, 3, 4 without repeating any one of them. What is the sum of all such possible numbers?

- a. 66660
- b. 25769
- c. 49180
- d. 76976

✓ **Verified Answer:** `A`

Problem 08 • Number Systems & Simplification

CORE SE

`ID: SEC02_BOOK_P14_Q99` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> Two dice are tossed. The probability that the total score is a prime number is:

- a. $\frac{5}{12}$
- b. $\frac{7}{12}$
- c. $\frac{9}{12}$
- d. $\frac{11}{12}$

✓ **Verified Answer:** `A`

Problem 09 • Number Systems & Simplification

CORE SE

ID: SEC02_BOOK_P18_Q131 | Source: 250 + Video Solution Quant Assorted Book.pdf | Level: Infosys SE Standard

> If $n = 1 + x$, where x is the product of four consecutive positive integers, then which of the following is/are true?
> A. n is odd
> B. n is prime
> C. n is a perfect square

- a. A and C only
- b. A and B only
- c. None of these
- d. A, B and C

✓ **Verified Answer:** `A`